Chang Hwan Kim

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Education

University of Michigan *Aug. 2026 – Present*
Ph.D. in Mechanical Engineering Ann Arbor, MI, USA
- Advisor: Prof. Gabor Orosz

Yonsei University *Sep. 2024 – Aug. 2026*
M.S. in Mechanical Engineering Seoul, Korea
- **GPA**: 4.23/4.30 (3.97/4.00)
- Advisor: Prof. Jun Young Yoon

B.S. in Mechanical Engineering *Mar. 2019 – Aug. 2024*
- **GPA**: 4.13/4.30 (3.92/4.00), ranked 1st out of 145 in major, **Summa Cum Laude**
- Includes a 1.5-year leave of absence for mandatory military service (Jan. 2021 – Jul. 2022)

Research Interests

Vehicle dynamics and control, model predictive control (MPC), control barrier functions (CBF), design and motion control of mechatronic systems

Publications

Journal Articles

J5. DNN-Approximated Offset-Free MPC on FPGA for 10-kHz Constrained In-Plane Control of 6-DOF Magnetic Levitation System
C.H. Kim, J.J. Kim, H.M. Yoon, J.Y. Yoon
IEEE Transactions on Industrial Informatics (**TII**), Early Access, 2026 [DOI] [PDF]

J4. Design and Control of Dual Reluctance-Lorentz 6-DOF Magnetic Levitator for In-Plane Long-Stroke Motion and Low-Power Vertical Levitation
C.H. Kim, H.M. Yoon, J.W. Jung, E.K. Kim, J.Y. Yoon
IEEE/ASME Transactions on Mechatronics (**TMECH**), under review (minor revision)

J3. Active 6-DOF Magnetic Levitator for Fully-Automated High-Throughput Large-Range Single-Mode Optical Fiber Alignment
H.M. Yoon, **C.H. Kim**, K.J. Kang, J.M. Sung, E.K. Kim, . . . , J.Y. Yoon
IEEE Transactions on Industrial Electronics (**TIE**) [PDF]

J2. Identification and Reduction of Normal-Direction Force Ripples in Permanent-Magnet Linear Synchronous Motors
S.M. Lee, Y.S. Kwon, J.H. Jung, K.J. Kang, **C.H. Kim**, J.Y. Yoon
IEEE Transactions on Industrial Electronics (**TIE**) [PDF]

J1. Study on Characteristics of SPGD Algorithm in Coherent Beam Combining with Phase and Tip-Tilt Control Speed Differences
H.M. Yoon, S.M. Lee, J.W. Jung, K. Lee, J.J. Jung, **C.H. Kim**, J.Y. Yoon
Journal of the Korean Society for Precision Engineering (**JKSPE**) [PDF]

Conference Presentations

C4. Control Barrier Function-Based Safety-Critical Control of 6-DOF Magnetic Levitation System
C.H. Kim, J. Kim, H.M. Yoon, J.Y. Yoon
Korean Society for Precision Engineering Spring Conference (**KSPE**), May 2026 [Abstract]

C3. Offset-Free Model Predictive Control of 6-DOF Magnetic Levitator for High-Precision Trajectory Tracking
C.H. Kim, J. Kim, H.M. Yoon, J.Y. Yoon
Korean Society for Precision Engineering Autumn Conference (**KSPE**), Nov. 2025 [Abstract]

C2. Gain-Scheduled Feedforward-Feedback Control for Reducing Lateral Acceleration and Jerk of Vehicle During Lane Changing Maneuver
C.H. Kim, J.K. Lee, H.S. Jang, J. Wang, J.Y. Yoon
International Conference on Precision Engineering and Sustainable Manufacturing (**PRESM 2025**), Jul. 2025 [Abstract]

C1. Control Method for Maintaining Dynamic Performance of a 6-DOF Hybrid-Reluctance-Type Magnetic Levitator During Payload Mass Changes
C.H. Kim, H.M. Yoon, J.W. Jung, J.Y. Yoon
International Conference on Precision Engineering and Sustainable Manufacturing (**PRESM 2024**), Jul. 2024, **Outstanding Presentation Award** [Abstract]

Research Experience

Yonsei University (Advisor: Prof. Jun Young Yoon) *Seoul, Korea*
Graduate Researcher *Sep. 2024 – Aug. 2026*
Undergraduate Researcher *Jan. 2023 – Aug. 2024*

FPGA-Based DNN-Approximated Offset-Free MPC for 6-DOF Magnetic Levitator *Mar. 2025 – Aug. 2026*
- Designed an offset-free model predictive controller (MPC) with a disturbance observer for constrained in-plane motion control (C3).
- Approximated the MPC law with a deep neural network (DNN) and implemented it on an FPGA, achieving constrained control at 10 kHz (J5).
- Developed control barrier function (CBF)-based safety-critical control for the same platform (C4).

Design and Control of 6-DOF Magnetic Levitator for Long-Stroke Zero-Power Gravity Compensation *Jan. 2024 – Jun. 2025*
- Designed and fabricated a 6-DOF magnetic levitator that combines reluctance forces for gravity compensation with Lorentz forces for long-stroke in-plane motion (J4).
- Developed and implemented 6-DOF feedback and feedforward controllers for precision motion control; the prototype levitates stably with only 0.498 W at rest.

Vehicle Lateral Dynamics Control for Reducing Lateral Acceleration and Jerk During Lane Changes *Sep. 2024 – Feb. 2025*
- Developed a gain-scheduled preview feedforward controller and integrated it with a lane-keeping LQR feedback controller (C2).
- Used a 2-DOF dual-track vehicle model with a linear tire model and Ackermann-geometry front-wheel steering, and designed a Luenberger-PI observer for state estimation.
- Reduced lateral acceleration, jerk, and lateral distance error by 36.4%, 79.6%, and 54.2% on average, respectively, relative to feedback-only control over 60–120 km/h in simulation.

Force Ripple Reduction Control of Permanent-Magnet Linear Synchronous Motor (PMLSM) *Mar. 2023 – Mar. 2024*
- Designed a DQ-axis current control method that reduces both normal- and tangential-direction force ripples, achieving 69% and 87% RMS reduction, respectively (J2).
- Proposed a method to identify the motor parameters that induce force ripples in iron-cored PMLSMs.

Adaptive Control of Low-Power 6-DOF Hybrid-Reluctance-Type Magnetic Levitator *Mar. 2023 – Dec. 2023*
- Designed a disturbance observer-based method to estimate the payload of the system.
- Implemented a controller that maintains the dynamic performance of the system under payload changes (C1, undergraduate thesis).

Industry-Sponsored Projects

Intelligent Controller Development for PAPS-Based Robotic C-Arm *Sep. 2025 – Aug. 2026*
In collaboration with GENORAY Co., Ltd.
- Designed sub-optimal collision-free paths and trajectories for a 4-DOF robotic C-arm using the A* algorithm.
- Implemented and tested the planners in a MATLAB simulation environment to ensure avoidance of the patient area.

Independent Drive and Control System for Styler Moving Hanger *Nov. 2024 – Oct. 2025*
In collaboration with LG Electronics Inc.

- Designed a novel drive mechanism enabling independent movement of the styler's moving hanger.
- Developed a motion controller that compensates for actuator cogging force and bearing friction.

Laser CBC SPGD Control Modeling and Simulation Development *Apr. 2024 – Mar. 2025*
In collaboration with LIG Nex1 Co., Ltd.

- Designed a stochastic parallel gradient descent (SPGD) algorithm for 19-channel laser coherent beam combining (J1).
- Developed a MATLAB simulator for laser beam modeling and SPGD performance evaluation.

Honors and Scholarships

Summa Cum Laude (Top 1%) *Aug. 2024*
Graduated with highest honors from Yonsei University; one of four recipients in the College of Engineering.

Outstanding Presentation Award *Jul. 2024*
Awarded at the International Conference on Precision Engineering and Sustainable Manufacturing (PRESM 2024).

Semester Honors, Yonsei University *Spring 2019, Fall 2019, Fall 2023, Spring 2024*

Accelerated B.S.–M.S. Program Scholarship, Yonsei University *Sep. 2024 – Aug. 2026*
Full tuition for 4 semesters, awarded for academic excellence upon early admission to the master's program.

Brain Korea 21 (BK21) Scholarship, National Research Foundation of Korea *Sep. 2024 – Aug. 2026*
Awarded to support graduate research in strategic fields under Korea's national research initiative.

Daesang Foundation Scholarship, Daesang Group *Mar. 2020 – Aug. 2024*
Full tuition for 6 semesters, awarded to outstanding undergraduate students.

Yonsei Eminence Scholarship, Yonsei University *Sep. 2019 – Feb. 2020*
One-third tuition for one semester, awarded for academic excellence.

Teaching Experience

MEU3014 – Mechatronics *Fall 2024*
Teaching Assistant, Yonsei University

- Topics: electronics, sensors, motors, linear control theory, and control practice using a microcontroller.

MEU3007 – Independent Study *Spring 2025*
Teaching Assistant, Yonsei University

- Supervised an undergraduate student on their undergraduate thesis.

Service and Other Experience

Republic of Korea Army *Jan. 2021 – Jul. 2022*
Infantry Rifle Squad Leader Inje, Korea

- Served in the Opposing Force (OPFOR) regiment, 2nd Mountain Regiment, Korea Combat Training Center (KCTC).

Korean Red Cross Blood Management Center *Jun. 2017 – Present*
Regular Blood Donor (20 donations since Jun. 2017) Seoul, Korea

Skills

Programming	MATLAB/Simulink, Python, C, LabVIEW, LaTeX
Toolboxes	MATLAB MPC, Control System, Optimization, Deep Learning, and Robotics Toolboxes; Vehicle Dynamics Blockset; MPT3
Design & Simulation	ANSYS Electronics, Autodesk Inventor
Languages	Korean (native), English (full professional proficiency)